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Prototype Memory-based Neighboring Feature Fusion Network for Image Manipulation Localization

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***Abstract*—Image manipulation localization (IML) aims to segment manipulated regions in suspicious images. However, most existing methods rely solely on intrinsic features extracted from the input image and passively model local or global inconsistencies, making it difficult to accurately delineate manipulated regions with ambiguous boundaries. To address these challenges, we propose a prototype memory-based neighboring feature fusion network (PNF-Net), which is inspired by a biological memory mechanism. PNF-Net simulates selective preference by learning manipulation-trace prototypes as memory priors, thereby guiding representation learning toward consistent and discriminative manipulation cues. Specifically, we propose a memory-guided localization module (MLM) that models the consistencies and anomalies between manipulated regions and the background as memory priors, enabling precise localization. We then propose a neighboring feature interaction module (NFIM) that preserves fine-grained details from neighboring shallow features, enhances global semantics from neighboring deep features, and effectively fuses them. Finally, a verification fusion module (VFM) is designed to enrich contextual semantics and improve the completeness and accuracy of localization results. Extensive experiments on multiple benchmark datasets show that our PNF-Net outperforms most state-of-the-art IML models. Our code is available on https://github.com/vpsg-research/PNF-Net.**

***Index Terms*—Image manipulation localization, prototype memory, neighboring feature, contextual semantics.**

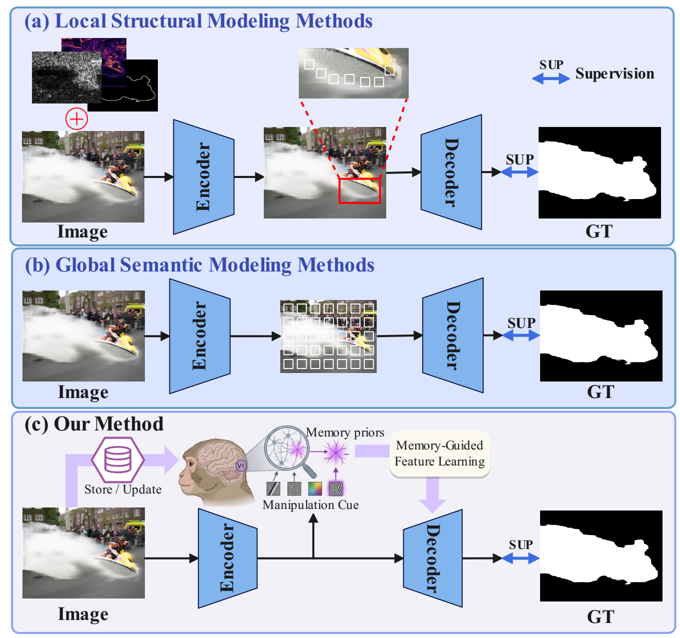

Fig. 1. Existing IML methods generally fall into two categories: (a) local structural modeling, which leverages boundary, noise, or frequency cues to capture local artifacts; and (b) global semantic modeling, which enhances semantic discrepancies between tampered regions and the background via network design and feature interactions. (c) We construct a clusterable memory bank that aggregates manipulation cues into reusable memory priors, which proactively guide representation learning.

I. Introduction

With the advancement of artificial intelligence, image manipulations have created highly realistic forgeries that are almost imperceptible to the human eye. The risk of maliciously altered image content poses significant challenges to academia, judiciary, military, and public safety. Therefore, developing a reliable image manipulation localization (IML) model that can accurately segment tampered regions has become an urgent necessity [1]–[3].

To address this problem, numerous deep learning based IML methods have been proposed. As illustrated in Fig. 1(a), some studies incorporate auxiliary cues, such as boundary maps [4]–[7], noise views [8]–[11], and frequency domain representations [12]–[14], to capture subtle manipulation traces that are difficult to perceive in the RGB domain, thereby strengthening local modeling. Meanwhile, as shown in Fig. 1(b), feature fusion methods [15]–[18] enhance global modeling by enriching contextual information and improving segmentation accuracy through carefully designed architectures. Despite these advances, most existing approaches remain largely input-driven: they passively mine local artifacts and global semantics, but lack an explicit mechanism to accumulate and reuse stable manipulation patterns across samples and scenes. Consequently,

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when traces are extremely subtle, boundaries are smoothed, manipulated regions are spatially discontinuous, or images undergo real-world degradations such as compression and resampling, local cues are often overwhelmed by noise and global semantics may produce seemingly plausible yet biased predictions. This can lead to overconfident false alarms, missed detections, and incomplete localization.

Inspired by the "grandmother cell" related neural mechanism in the primary visual cortex (V1) of macaques [19], we incorporate this biological principle into the IML task. Prior studies suggest that such neurons can be regarded as storage units for complex visual prototypes. When an incoming stimulus closely matches a neuron's preferred pattern, its response can be 3 to 5 times stronger than that elicited by a generic stimulus. By analogy, we argue that an ideal IML model should not rely solely on passively extracted features from the input image. Under degradations such as compression and noise, a model that predicts purely from corrupted features is inherently constrained. To address this issue, we build a memory bank via clustering to emulate selective neural responses, as illustrated in Fig. 1(c). Functionally analogous to grandmother cells, the memory bank aggregates dispersed manipulation cues observed during training (e.g., inter-region inconsistencies and anomalous artifacts) into compact, reusable memory priors. During inference, these priors serve as reference templates for matching and retrieval. Even when the input image is degraded, if a local feature aligns well with a stored prototype, the model retrieves and activates the corresponding memory representation to strengthen the response.

Based on the above considerations, we propose a novel IML framework called prototype memory-based neighboring feature fusion network (PNF-Net) for accurately localizing tampered regions in images. First, we propose a memory-guided localization module (MLM), which leverages semantic clusters to accumulate the consistencies and anomalous features between tampered regions and the background within images. These form memory priors that guide the IML representation learning, enhancing the accuracy of locating tampered regions. Then, we propose a neighboring feature interaction module (NFIM). NFIM fuses features of different scales through a series of sequential dilated convolutions while utilizing skip connections to generate more robust and effective feature representations. Additionally, NFIM adopts a dual-branch cross-refinement strategy for neighboring shallow features, significantly preserving detail information. For neighboring deep features, NFIM combines contextual information and captures global semantic information through feature weighting. Given the importance of both global semantic information and detail information for accurately localizing and segmenting tampered regions, NFIM enhances IML's feature representation learning ability by integrating neighboring features. To further enrich contextual semantic information and improve the completeness and accuracy of segmentation results, we propose a verification fusion module (VFM). VFM establishes channel connections between higher and lower-level features through weight maps, achieving more effective multi-level feature fusion. The integrity of the feature masks output by VFM at different levels is verified through a multi-level supervision strategy, ultimately generating refined masks of tampered regions. The main contributions of this paper are threefold:

- We propose a novel PNF-Net for IML, which uses memory priors to guide representation learning for precise localization of tampered regions. It effectively integrates valuable information from neighboring features to form robust fused feature representations. Extensive experiments show that our model achieves SOTA performance.
- We propose an MLM inspired by "grandmother cells" in the primary visual cortex (V1) of macaque monkeys. The MLM uses semantic clusters to accumulate consistencies and anomalies between tampered regions and the background in images, forming memory priors that guide IML representation learning.
- We design a hierarchically collaborative feature refinement mechanism comprising the NFIM and VFM. NFIM aggregates adjacent complementary cues to handle scale variations, while VFM establishes cross-level channel dependencies and enforces structural integrity via multi-level supervision. This synergy enables robust representations for localizing tampering artifacts.

II. Related Work

A. Image Manipulation Localization

Image manipulation localization (IML) aims to segment tampered regions in suspicious images. With the rapid development of deep learning, several methods have achieved remarkable performance. Chen *et al.* [21] improved the BusterNet for image copy-move forgery detection, addressing its limitations in region localization accuracy and feature extraction resolution. Shi *et al.* [22] introduced TA-Net, leveraging multi-scale transformers and boundary information to improve segmentation accuracy. Guo *et al.* [23] proposed AdapGRnet, which introduces a fine-grained manipulation trace extractor (MTE) and an attention fusion mechanism (AFM) to adaptively and synergistically fuse spatial- and residual-domain features. Xu *et al.* [24] proposed a novel method that utilizes residual attention and integrates multi-scale local and global information to enhance detection accuracy. Xiong *et al.* [25] proposed CMCF-Net, which effectively fuses contextual and multiscale information in a weighted manner, thereby enhancing the ability to localize tampering traces at different scales. Weng *et al.* [26] proposed a UCM-Net, which employs multiple asymmetric cross-layer connections, self-correlation computation, and atrous spatial pyramid pooling to differentially handle tampered regions of varying sizes. Li *et al.* [27] proposed SCAF, which integrates structural consistency, prior-aware feature modulation, and gated adaptive fusion modules, substantially improving the model's robustness and localization accuracy. Guo *et al.* [28] leveraged SAM with coarse prompts to generate low-cost, fine-grained pseudo masks for supervising student model training, thereby enhancing generalization and localization accuracy in complex manipulation scenarios.

Most existing IML methods simply fuse features in a bottom-up manner, which often leads to the propagation of

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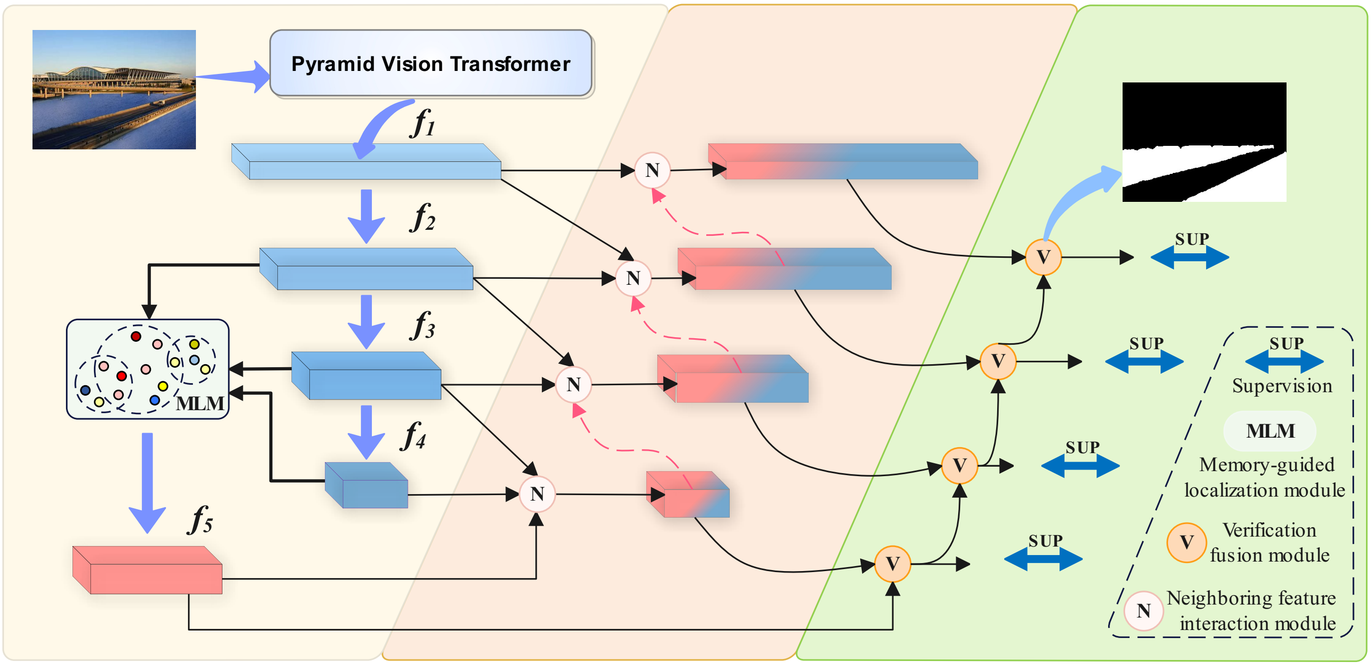

Fig. 2. The overall architecture of PNF-Net. It utilizes PVTv2 [20] as the backbone and incorporates three main modules, namely, the memory-guided localization module (MLM), the neighboring feature interaction module (NFIM), and the verification fusion module (VFM).

irrelevant semantic features, causing confusion between the target and the background. This means that some semantic features are introduced into shallow features, making it difficult to fully represent detailed information. The same issue applies to deep features. Our key innovation is twofold. First, we construct a memory bank that stores prototypes of typical manipulation patterns and retrieve/match these prototypes during feature fusion to memory-guidedly regulate information flow, thereby highlighting manipulation-relevant cues and suppressing background noise. Second, motivated by the distinct representational tendencies of features at different network depths, we selectively distill and reorganize fine-grained details from shallow layers and high-level semantics from deep layers, and integrate them in a principled cross-level fusion manner. This complementary integration yields manipulation-localization representations that are semantically informative while preserving fine structural details.

B. Prototype Memory Mechanism

The prototype memory bank can capture diverse features over extended periods and has shown effectiveness in multiple domains. Zhang *et al.* [29] introduced a memory unit to store task-specific information, enabling the model to adapt to specific tasks. Zhu *et al.* [30] designed an invariant feature storage module to store domain-level prototype information, enhancing representation and improving semantic segmentation accuracy in high-resolution remote sensing images. Wang *et al.* [31] established a multi-scale prototype memory queue to address scene variation and scale differences in high-resolution aerial images. Lee *et al.* [32] adaptively stored useful prototypes to enhance relationships between distant frames, improving unsupervised video object segmentation accuracy.

Existing prototype memory techniques store images at the instance level for object recognition, but IML is a binary segmentation task with only target and background categories, making memory bank construction overly limited. As a result, prototype memory techniques have not yet been introduced to the IML field. In this paper, we move beyond constructing a memory bank solely for tampered regions and the background. Instead, we address various tampering types, such as splicing, copy-move, and inpainting. Using momentum online clustering, we learn consistencies and anomalies across different tampering types, as well as between tampered regions and the background, effectively applying prototype memory techniques to the IML field.

III. Methodology

A. Overview

The structure of our proposed PNF-Net is shown in Fig. 2. We utilize PVTv2 [20] as the backbone network. Initially, the backbone network is used to extract multi-level features from the input image, denoted as $\boldsymbol{f}_i, i \in \{1, \ldots, 4\}$. Then, we pass $\boldsymbol{f}_2$, $\boldsymbol{f}_3$, and $\boldsymbol{f}_4$ through a memory-guided localization module (MLM) to generate representations with higher-level semantics, denoted as $\boldsymbol{f}_5$. Next, we design multiple neighboring feature interaction modules (NFIM), which generate fused features $\boldsymbol{F}_i$ by integrating valuable information from neighboring features. Finally, the verification fusion module (VFM) enriches the contextual information to generate precise prediction masks.

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B. Memory-guided Localization Module

The global perspective provided by deep semantics plays a crucial role in localizing tampered regions in images, but relying solely on this perspective may lead to overconfident incorrect predictions. To address this issue, we draw inspiration from the concept of "grandmother cells" in the primary visual cortex (V1) of macaque monkeys. Research has shown that "grandmother cells" are highly selective, responding strongly to specific stimuli while remaining largely unresponsive to irrelevant ones. In the context of image manipulation localization (IML), we simulate this selective feature, enabling the model to react strongly to tampering traces (e.g., edges or textures of tampered regions) while ignoring irrelevant background information. Building on this idea, we propose a memory-guided localization module (MLM) that learns the consistencies and differences between various tampering traces and the background to build a prototype memory bank. In this memory bank, the features of each tampering trace are highly specific, much like "grandmother cells." When localizing tampered regions, the MLM selects the most similar prototype features as memory priors to guide the model in precisely locating tampered areas. This process allows the model to focus more accurately on tampering traces rather than background noise or unrelated areas, thereby improving localization accuracy. As shown in Fig. 3, we concatenate the features from the last three stages of the backbone network ($\boldsymbol{f}_2$, $\boldsymbol{f}_3$, $\boldsymbol{f}_4$) to obtain a higher-level semantic representation $\boldsymbol{c}$. This can be expressed as follows:

$$\boldsymbol{c} = Concat\left(D(\boldsymbol{f}_2), \boldsymbol{f}_3, U(\boldsymbol{f}_4)\right) \tag{1}$$

Where $Concat$ denotes the concatenation operation. D denotes down-sampling. U denotes up-sampling. Subsequently, $\boldsymbol{c}$ is transformed into queries, keys, and values using corresponding 1×1 convolutions $Q(\cdot)$, $K(\cdot)$, and $V(\cdot)$. Then, we construct a prototype memory bank (M). Formally, we assume M contains n semantic units $M = \{\boldsymbol{m}_1, \boldsymbol{m}_2, \ldots, \boldsymbol{m}_n\}$. Each semantic unit m_j consists of l prototype units $\{\boldsymbol{e}_1^{(j)}, \boldsymbol{e}_2^{(j)}, \ldots, \boldsymbol{e}_l^{(j)}\}$, and each semantic unit represents the mean of its associated prototype units' clusters. These prototype units are updated through the feature maps from previous iterations, transformed by a momentum-based function $K^m(\cdot)$, which is the momentum counterpart of $K(\cdot)$. The parameter update process through the momentum parameter x is as follows:

$$K^m \leftarrow K^m \times (1 - x) + K \times x \tag{2}$$

After being processed by $K^m(\cdot)$, the feature map replaces the prototype unit in the most similar semantic cluster. A random replacement strategy is employed to prevent all prototype units from collapsing into trivial solutions. Then we select the prototype unit $\boldsymbol{e}^{(j)}$ from the most similar semantic unit as memory prior to guide IML representation learning, resulting in the feature $\boldsymbol{A_m}$ that possesses prototype characteristics. The details are as follows:

$$\boldsymbol{A_m} = Softmax(Q(\boldsymbol{c}) \times \boldsymbol{e}^{(j)}) \times \boldsymbol{e}^{(j)} \tag{3}$$

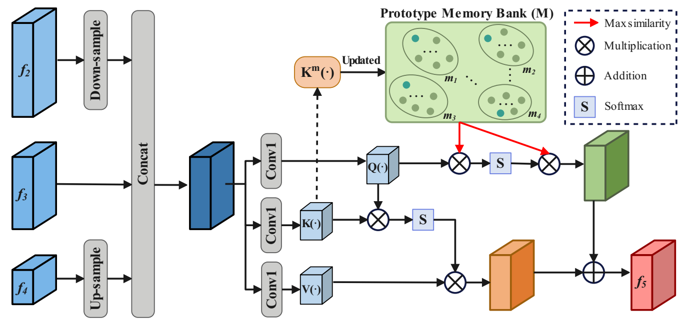

Fig. 3. The architecture of memory-guided localization module.

A global perspective is also crucial for localizing tampering traces in images. We capture global context information using self-attention, resulting in $\boldsymbol{A_s}$. Finally, we sum $\boldsymbol{A_s}$ with $\boldsymbol{A_m}$ and map it back to the original space through a 1×1 convolution to obtain the feature $\boldsymbol{f_5}$, thereby achieving the modulation of memory and global information.

$$\begin{cases} \boldsymbol{A}_s = Softmax(Q(\boldsymbol{c}) \times K(\boldsymbol{c})) \times V(\boldsymbol{c}) \\ \boldsymbol{f}_5 = \mathrm{Conv}\,1(\boldsymbol{A}_m + \boldsymbol{A}_s) \end{cases} \tag{4}$$

Where $\mathrm{Conv}k$ denotes $k \times k$ convolution. By using memory priors, the model can more accurately identify anomalies and consistent features within images. This helps to reduce false positives and false negatives, thereby improving the precision of tampered region localization.

C. Neighboring Feature Interaction Module

According to [33], low-level features in shallow layers contain rich details, while high-level features in deep layers hold global semantic information. Merging these features without addressing their distinct characteristics may introduce interference, leading to confusion between the target and background. This results in semantic features being introduced into shallow low-level features, hindering the representation of detailed information, and vice versa. Our goal is to complement the detail and semantic information at different levels, producing representations that are rich in semantic information while preserving detailed features.

In response to these issues, we propose a neighboring feature interaction module (NFIM), as illustrated in Fig. 4. It consists of three components: a multi-scale information module (MSIM), a detail preservation module (DPM), and a semantic enhancement module (SEM). The current stage features are processed through the MSIM to extract multi-scale information. The DPM is designed to mine detail information from neighboring shallow features, while the SEM strengthens the semantic information of neighboring deep high-level features. Ultimately, these features are fused together to produce a robust feature representation.

1) Multi-scale information module (MSIM): Different sizes of receptive fields are crucial for image manipulation localization, but serial use of convolutional operations fails to extract rich contextual information. Therefore, we employ a series of dilated convolutions to achieve larger receptive fields and supplement information through skip connections.

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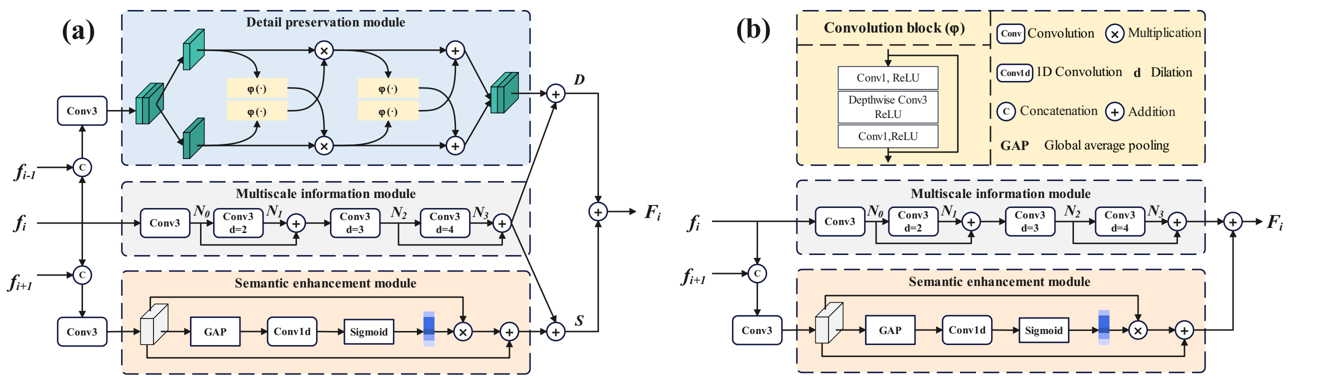

Fig. 4. The architecture of neighboring feature interaction module. It is worth noting that the processing of $\boldsymbol{f}_i, (i = 2, \ldots, 4)$ is shown in Figure (a), while the processing of $\boldsymbol{f}_1$ is shown in Figure (b).

Specifically, we input the i-th level feature $\boldsymbol{f}_i$ into the MSIM to mine multi-scale information. The MSIM includes one 3×3 convolution layer and three 3×3 dilated convolution layers with dilation rates of 2, 3, and 4, respectively. Subsequently, we use skip connections to supplement the features. This approach enables better utilization of features at different scales and helps prevent the loss of information within the network. It can be formulated as:

$$\boldsymbol{N}_k = \begin{cases} \text{Conv}\,3\,(\boldsymbol{f}_i)\,, & k=0 \\ F_{\text{Conv3}}^{d=k+1}\,(\boldsymbol{N}_{k-1})\,, & k=1,3 \\ F_{\text{Conv3}}^{d=k+1}\,(\boldsymbol{N}_{k-1}+\boldsymbol{N}_{k-2})\,, & k=2 \end{cases} \tag{5}$$

Where $F_{\text{Conv3}}^{d=k+1}$ denotes a 3×3 dilated convolution layer with a specific dilation rate d. Finally, $\boldsymbol{N}_2$ and $\boldsymbol{N}_3$ are added element-wise and processed through a 1×1 convolution to obtain the multi-scale feature $\boldsymbol{X}_i$:

$$\boldsymbol{X}_i = \text{Conv}\,1\,(\boldsymbol{N}_2+\boldsymbol{N}_3) \tag{6}$$

MSIM captures broad context by increasing the receptive field, enhancing the network's ability to perceive tampered regions. Skip connections improve segmentation by facilitating multi-scale information exchange.

2) Detail preservation module (DPM): Detail features are crucial for segmenting structurally complete regions. However, as networks deepen, detail information is gradually lost. We propose a DPM to extract and retain detail information from shallow neural networks. This module supplements detailed information into neighboring deep high-level features, enhancing detail representation. Specifically, we concatenate $\boldsymbol{f}_i$ with the shallower feature $\boldsymbol{f}_{i-1}$, then split them into two parts ($\boldsymbol{q}_1$, $\boldsymbol{q}_2$), which are processed in parallel through different convolutional blocks $\varphi(\cdot)$. Finally, the results of the two branches are multiplied element-wise to obtain $\boldsymbol{Y}_1$ and $\boldsymbol{Y}_2$. The detailed process is as follows:

$$\begin{cases} \boldsymbol{Y}_1 = \varphi\,(\boldsymbol{q}_1) \times \boldsymbol{q}_2 \\ \boldsymbol{Y}_2 = \varphi\,(\boldsymbol{q}_2) \times \boldsymbol{q}_1 \end{cases} \tag{7}$$

Where $\varphi(\cdot)$ denotes a 1×1 convolution layer followed by a depthwise separable convolution. In this context, the fused feature maps capture the correlations between different channels. Subsequently, $\boldsymbol{Y}_1$ and $\boldsymbol{Y}_2$ are re-input into $\varphi(\cdot)$ for feature enhancement. To retain more detail information, the enhanced features are cross-added with the original branches. Finally, we concatenate $\boldsymbol{Y}_1'$ and $\boldsymbol{Y}_2'$ to obtain the detail feature L. This process is described as follows:

$$\begin{cases} \boldsymbol{Y}_1' = \varphi\,(\boldsymbol{Y}_1) + \boldsymbol{Y}_2 \\ \boldsymbol{Y}_2' = \varphi\,(\boldsymbol{Y}_2) + \boldsymbol{Y}_1 \\ \boldsymbol{L} = (Concat\,(\boldsymbol{Y}_1', \boldsymbol{Y}_2')) \end{cases} \tag{8}$$

DPM divides the fused features into two maps along the channel and processes them in parallel. It then uses depthwise separable convolution to retain spatial structure information. The two branches are cross-multiplied and added element-wise to enhance channel correlations, capturing richer details. Finally, multi-scale feature $\boldsymbol{X}_i$ and $\boldsymbol{L}$ are added element-wise to obtain $\boldsymbol{D}$.

$$\boldsymbol{D} = \boldsymbol{L} + \boldsymbol{X}_i \tag{9}$$

DPM extracts detail information from neighboring shallow features and injects it into $\boldsymbol{X}_i$, enhancing the feature representation with object structure details.

3) Semantic enhancement module (SEM): Global semantic features are crucial for localizing tampered regions in images. To enhance semantic information and eliminate shallow detail interference, we propose a SEM to suppress irrelevant semantics and extract valuable global features. We first concatenate $\boldsymbol{f}_i$ and $\boldsymbol{f}_{i+1}$ to obtain the fused feature $\boldsymbol{H}$. Global average pooling (GAP) integrates spatial information in $\boldsymbol{H}$ into a global feature, capturing the overall characteristics of tampered regions. The $1D$ convolution followed by a sigmoid function generates the channel weighting mask $\boldsymbol{E}$, which is multiplied element-wise with the backbone features to enhance global semantics. Residual connections preserve the original features. Finally, a 1×1 convolution is used for feature fusion to obtain $\boldsymbol{H}'$. It can be formulated as:

$$\begin{cases} \boldsymbol{E} = \text{Conv1}\left(\sigma\left(W_{1D}^{k}(\text{GAP}(\boldsymbol{H}))\right)\right) \\ \boldsymbol{H}' = \text{Conv1}(\boldsymbol{H} \times \boldsymbol{E} + \boldsymbol{H}) \end{cases} \tag{10}$$

Where σ denotes sigmoid function. W_{1D}^{k} denotes a 1D convolution with a kernel size of k. We set $t = \lfloor|\,(\log_2(C)+1)\,|\,/2\rfloor$, where C is the number of channels. $\lfloor\ \rfloor$ denotes rounding down. If t is even, then $k = t+1$; if t is odd, then $k = t$. Subsequently, we perform element-

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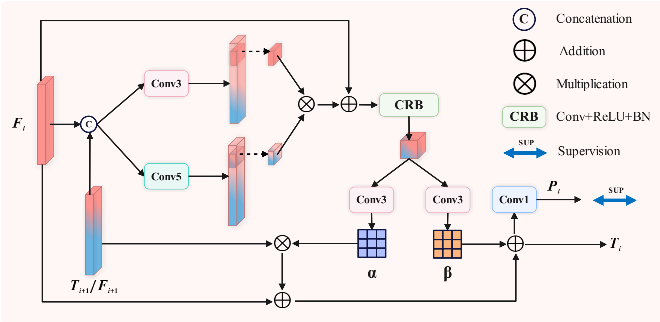

Fig. 5. The architecture of verification fusion module.

wise addition of $\boldsymbol{H}'$ and $\boldsymbol{X}_i$ to augment the global semantic information in $\boldsymbol{X}_i$.

$$\boldsymbol{S} = \boldsymbol{H}' + \boldsymbol{X}_i \tag{11}$$

To generate features rich in semantic information while retaining more detailed feature representations, we perform element-wise addition of $\boldsymbol{S}$ and $\boldsymbol{D}$ to obtain $\boldsymbol{F}_i$.

$$\boldsymbol{F}_i = \boldsymbol{S} + \boldsymbol{D} \tag{12}$$

The NFIM preserves detail features from the shallow layer $\boldsymbol{f}_{i-1}$ and enhances semantic features from the deep layer $\boldsymbol{f}_{i+1}$, integrating them into $\boldsymbol{f}_i$. This helps the network compensate for the lack of semantic features in shallow layers and the loss of detail in deep layers.

D. Verification Fusion Module

It is well known that the fusion of contextual features is crucial for restoring the integrity of manipulated image regions. To improve the completeness and accuracy of the results, we propose a verification fusion module (VFM), as illustrated in Fig. 5. The VFM establishes connections between hierarchical features through weight maps, fusing features from different levels and extracting rich contextual information. A multi-level supervision strategy is used to verify the completeness of the feature masks output by the VFM at different levels. Specifically, using $\boldsymbol{F}_i$ and $\boldsymbol{F}_{i+1}$ as examples, we first concatenate them to obtain the fused feature $\boldsymbol{F}_m$. Next, we apply both a 3×3 convolution and a 5×5 convolution to $\boldsymbol{F}_m$, and then perform element-wise multiplication of the results to obtain $\boldsymbol{F}_i'$. Finally, we use residual connections to preserve the original features of $\boldsymbol{F}_i$, resulting in the output feature $\boldsymbol{O}$. It can be formulated as:

$$\begin{cases} \boldsymbol{F}_i' = \mathrm{Conv}\,3\,(\boldsymbol{F}_m) \times \mathrm{Conv}\,5\,(\boldsymbol{F}_m) \\ \boldsymbol{O} = CRB\,(\boldsymbol{F}_i' + \boldsymbol{F}_i) \end{cases} \tag{13}$$

Where CRB denotes a sequence of applying a 3×3 convolution layer, a ReLU function, and batch normalization. By using two parallel convolutions with different kernel sizes, we capture information at different scales and identify feature correlations. To achieve effective cross-layer feature interaction and enrich contextual semantic information, we generate two weight maps, α and β, by two 3×3 convolution layers. Subsequently, the weight maps establish cascading relationships between hierarchical features. By applying channel attention to features at different levels, valuable contextual information can be fully propagated to the decoder, achieving effective integration of feature maps across different layers. It can be defined as

$$\boldsymbol{T}_i = \alpha \times \boldsymbol{F}_{i+1} + \boldsymbol{F}_i + \beta \tag{14}$$

$\boldsymbol{T}_i$ is the output of the VFM. For $i = \{1, \ldots, 3\}$, the previous VFM output ($\boldsymbol{T}_{i+1}$) and $\boldsymbol{F}_i$ are used as inputs for the next VFM. The 1×1 convolution is then applied to $\boldsymbol{T}_i$ to generate prediction maps $\boldsymbol{P}_i$ for manipulated image regions at different levels. Finally, we implement a multi-level supervision strategy by comparing the VFM output at each stage with the ground truth to compute the loss. This ensures the integrity of the predicted masks at every stage.

TABLE I
EXPERIMENTAL DATASETS. MANIPULATION TYPES: COPY–MOVE (CM), SPLICING (SP), AND INPAINTING (IP).

Dataset	Nums	#CM	#SP	#IP	Train	Test
CASIAv2	5123	3295	1828	0	5123	0
NIST16	564	68	288	208	383	181
Coverage	100	100	0	0	70	30
CASIAv1	920	459	461	0	0	920
Columbia	180	0	180	0	0	180
IMD2020	2010	–	–	–	0	2010
Korus	220	–	–	–	0	220

E. Overall Loss Function

The VFM produces four pseudo-target masks, denoted as $\boldsymbol{P}_i$ ($i \in \{1, \ldots, 4\}$). Since manipulated regions usually exhibit irregular structures and ambiguous boundaries, we adopt the pixel position aware loss (L_{ppa}) proposed by Wei *et al.* [34] as the supervision objective. This loss consists of a weighted binary cross-entropy term and a weighted IoU term:

$$L_{ppa} = L_{BCE}^{w} + L_{IOU}^{w} \tag{15}$$

where the pixel-wise weighting strategy in [34] assigns larger weights to hard pixels, especially boundary and noisy regions. This loss is particularly suitable for PNF-Net for two reasons. First, the weighted BCE term improves pixel-level discrimination for tampered and non-tampered regions, which is consistent with the objective of the proposed memory-guided localization module to suppress background interference. Second, the weighted IoU term enhances region-level structural consistency and boundary completeness, which complements the neighboring feature interaction and verification fusion processes for recovering fine-grained tampering traces. Therefore, the adopted loss helps optimize both global region integrity and local boundary quality.

The total loss is computed by summing the loss over the four output scales:

$$L_{all} = \sum_{i=1}^{4} L_{ppa}(\boldsymbol{P}_i, \boldsymbol{G}) \tag{16}$$

where $\boldsymbol{G}$ denotes the ground truth mask.

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TABLE II
PERFORMANCE COMPARISON WITH OTHER ADVANCED IML METHODS IN TERMS OF F1 SCORE (FIXED THRESHOLD AT 0.5). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED.

Method	Pub.	Params	FLOPs	In-distribution (ID)				Out-of-distribution (OOD)			
				CASIAv1	Coverage	NIST16	**Avg.**	Columbia	Korus	IMD2020	**Avg.**
PSCC-Net [16]	TCSVT'22	3.67M	29.06G	0.485	0.396	0.300	0.394	0.659	<u>0.232</u>	0.331	<u>0.407</u>
Trufor [35]	CVPR'23	68.70M	221.83G	0.523	0.423	0.373	0.440	<u>0.668</u>	0.204	0.333	0.402
MFI-Net [17]	TCSVT'24	32.54M	36.25G	0.670	0.440	0.454	0.521	0.524	0.157	0.284	0.322
Mesorch [36]	AAAI'25	85.75M	124.93G	<u>0.706</u>	<u>0.588</u>	0.540	<u>0.611</u>	0.550	0.206	<u>0.334</u>	0.363
SparseViT [37]	AAAI'25	50.30M	46.20G	0.483	0.435	**0.653**	0.524	0.409	0.156	0.309	0.291
PIM [38]	TPAMI'25	152.47M	682.88G	0.505	0.464	0.260	0.410	0.596	0.134	0.272	0.334
Ours	-	67.56M	68.60G	**0.754**	**0.737**	<u>0.588</u>	**0.693**	**0.738**	**0.234**	**0.445**	**0.472**

IV. EXPERIMENTS AND RESULTS

A. Datasets and Processing

Our experiments utilize 7 mainstream benchmark datasets: CASIAv1 [39], CASIAv2 [39], NIST16 [40], Columbia [41], Coverage [42], IMD2020 [43], and Korus [44]. Specifically, CASIAv1 and CASIAv2 serve as representative tampering datasets, featuring splicing and copy-move manipulations across diverse image contents and post-processing conditions. NIST16 provides more challenging and realistic manipulated images, characterized by complex tampering scenarios and corresponding localization annotations. Columbia is a foundational and widely adopted splicing dataset, primarily utilized to evaluate the detection of manipulation boundaries. Coverage focuses on copy-move forgeries with similar foreground and background regions, significantly increasing the difficulty of manipulation localization. IMD2020 contains high-quality manipulated images with pixel-level ground truth masks, making it ideal for evaluating fine-grained localization performance. Korus is designed to assess model generalization across diverse and highly challenging real-world manipulations. By evaluating on these datasets, we comprehensively demonstrate the effectiveness and robustness of the proposed method across various manipulation types and data distributions.

The specific partitions of each dataset are detailed in Table I. It is worth noting that NIST16 was processed using the method proposed by Ma *et al.* [45].

B. Performance Metrics

Image manipulation localization is formulated as a pixel-wise binary classification problem. We adopt the pixel-level F_1 score to measure the discrepancy between the ground-truth mask and the predicted mask. The F_1 score is defined as the harmonic mean of precision and recall. Although the Area Under the Curve (AUC) is another widely adopted evaluation metric in the IML field, Ma *et al.* [45] demonstrated that the AUC metric tends to yield overly optimistic estimations for IML tasks. Therefore, to more objectively reflect the true performance of the models, we exclusively utilize the F_1 score for evaluation in this paper. For fair comparison, we fix the threshold at 0.5. The $\boldsymbol{F_1}$ score can be computed as follows:

$$\boldsymbol{F_1} = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{17}$$

C. Implementation Details

In this work, we use PVTv2 [20] pre-trained on ImageNet as the backbone. During training, all images were resized to 416×416. The model was trained with a batch size of 32 using the AdamW optimizer [46]. The initial learning rate was set to 1e-4 and decayed by a factor of 0.1 every 50 epochs. Training was performed on 2 NVIDIA 3090 GPUs, with convergence reached after 60 epochs.

D. Comparison with SOTA Methods

Image manipulation localization: As reported in Table II, our proposed model achieves the best overall F1 performance among the compared advanced IML approaches on both in-distribution and out-of-distribution benchmarks. This superiority is mainly attributed to the design of prototype memory priors inspired by the "grandmother cells" phenomenon: we simulate the formation of stimulus-preferred responses by learning tampering-trace–aware prototypes, which serve as memory priors to guide IML representation learning. Concretely, the memory-guided localization module (MLM) captures the consistencies and anomalies between tampered regions and the background and leverages the most relevant prototypes to enhance the sensitivity to subtle manipulation traces, leading to more accurate segmentation of tampered regions. Moreover, considering that shallow and deep features exhibit different representational tendencies, PNF-Net adopts level-aware processing to mitigate the propagation of irrelevant semantics: neighboring shallow features are encouraged to preserve boundary details, while neighboring deep features are enhanced with stronger global semantics. Based on this principle, the neighboring feature interaction module (NFIM) effectively integrates multi-level information, and the verification fusion module (VFM) further enriches contextual semantics to improve the completeness and accuracy of IML.

Visual comparison: To demonstrate our model's strong ability to jointly exploit local and global information, Fig. 6 presents segmentation results for three scenarios: multi-target tampering (rows 1–2), large-scale tampering (rows 3–4), and small-scale tampering (rows 5–6). For small-scale and multi-target tampering, PNF-Net precisely localizes manipulated regions, whereas competing methods yield suboptimal results

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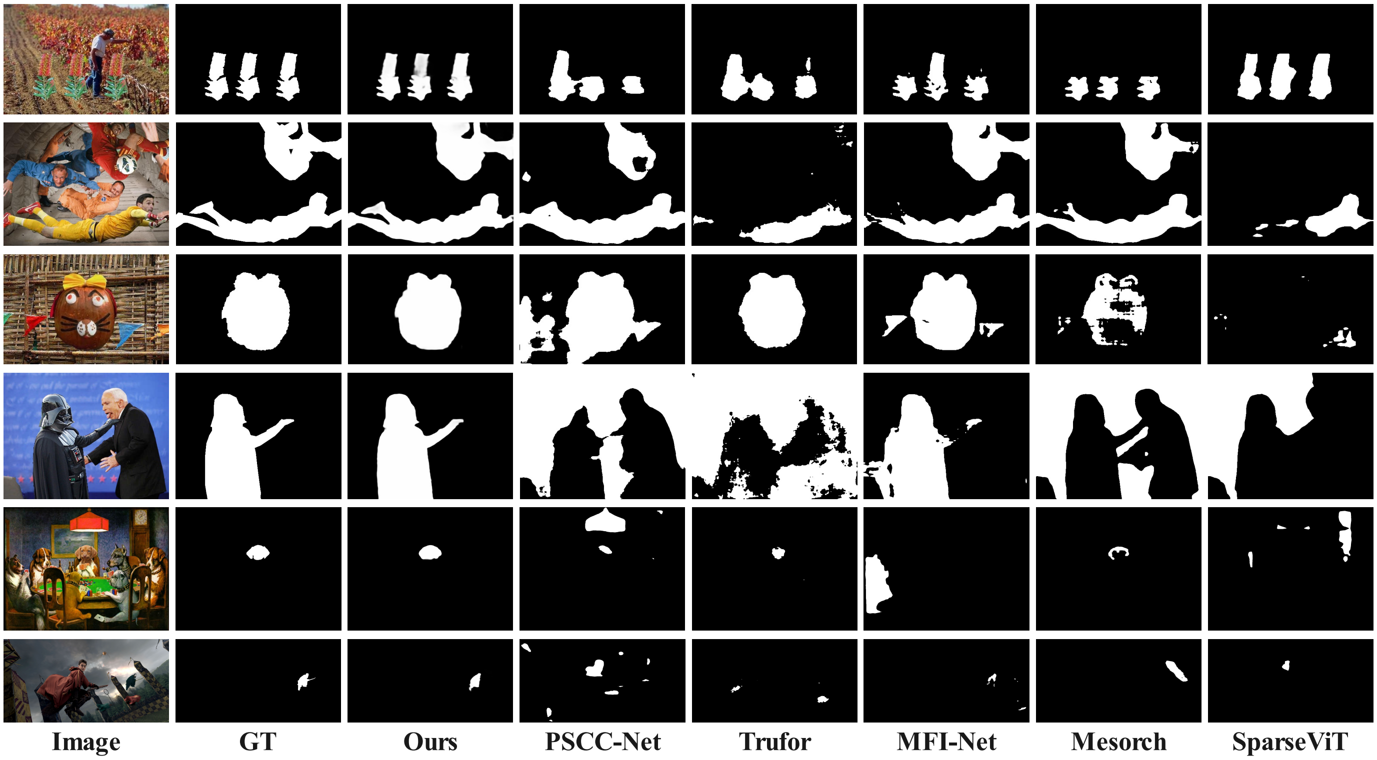

Fig. 6. We selected three challenging scenarios, namely small-scale tampering, large-scale tampering, and multi-target tampering, to demonstrate the capability of the proposed PNF-Net to jointly model local and global features.

TABLE III
ABLATION STUDIES ON EACH IN-DISTRIBUTION (ID) AND OUT-OF-DISTRIBUTION (OOD) DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD.

No.	Method	In-Distribution (ID)				Out-of-Distribution (OOD)			
		NIST16	CASIAv1	Coverage	Avg.	Columbia	Korus	IMD2020	Avg.
(a)	Ours w/o MLM	0.584	0.734	0.724	0.681	0.736	0.230	0.435	0.467
(b)	Ours w/o NFIM	0.583	0.747	0.724	0.685	0.728	**0.240**	0.433	0.467
(c)	Ours w/o VFM	0.584	0.722	0.723	0.676	0.711	0.193	0.416	0.440
(d)	Ours (full model)	**0.588**	**0.754**	**0.737**	**0.693**	**0.738**	0.234	**0.445**	**0.472**

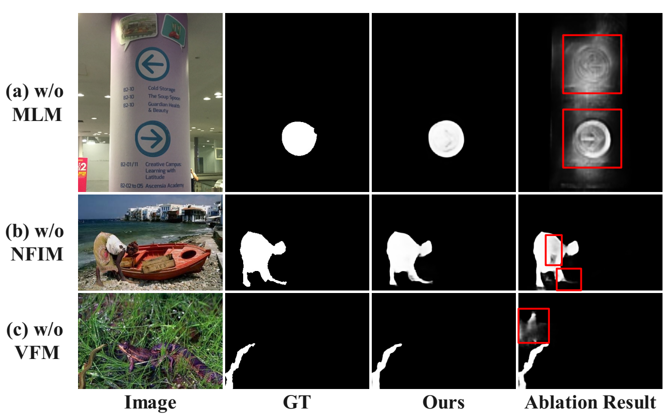

Fig. 7. Visualization of the ablation study. The red boxes highlight specific failure cases of the ablated models, contrasting with the accurate predictions of our full model.

or even fail. This advantage mainly stems from the carefully designed MLM, which effectively analyzes tampering traces and robustly captures deep semantic cues. Consequently, PNF-Net reduces missed detections and accurately identifies subtle tampering artifacts. For large-scale tampering, our model produces finer prediction masks than other methods because it leverages shallow features to complement deep features with detailed spatial information, enabling better detection of local artifacts and more refined masks. Overall, across all scenarios, our method consistently outperforms the compared models and yields predictions that better align with the ground truth.

E. Ablation Study

As shown in Table III, we conduct ablation experiments to validate the effectiveness of each key component in PNF-Net, including the memory-guided localization module (MLM), the neighboring feature interaction module (NFIM), and the verification fusion module (VFM). Overall, removing any module consistently degrades performance, indicating that

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TABLE IV
ABLATION STUDIES ON EACH COMPONENT OF NFIM.

No.	Method	Avg.ID	Avg.OOD
(a)	NFIM w/o MSIM	0.683	0.452
(b)	NFIM w/o DPM	0.669	0.460
(c)	NFIM w/o SEM	0.662	0.462
(d)	Ours (full model)	0.693	0.472

the proposed design is not a collection of isolated tricks but a cohesive framework in which the three components provide complementary benefits for robust IML. Furthermore, in conjunction with the qualitative visualization in Fig. 7, the specific contributions of each module to the final localization performance can be observed more intuitively.

Effectiveness of MLM. The MLM is designed to simulate a prototype-memory prior that encourages the model to develop a preference for tampering traces, rather than being dominated by content semantics. By explicitly modeling consistencies and anomalies between tampered regions and the background, MLM provides a stable prior for representation learning. Without MLM, the model relies purely on data-driven feature learning, which is more susceptible to weak and sparse tampering cues and tends to be distracted by irrelevant appearance variations. As shown in Table III, the performance degradation observed after removing the MLM proves that the prototype-memory prior is crucial for enhancing the model's sensitivity to subtle tampering artifacts and improving its generalization capability under distribution shifts. Furthermore, this can be observed more intuitively in conjunction with the qualitative visualization in Fig. 7(a). In the ablation result without the MLM, the model produces severe false alarms by incorrectly predicting the untampered top sign as a manipulated region (highlighted by the red boxes). Because the top and bottom signs are highly similar in visual appearance and content semantics, the model lacking the guidance of prototype memory priors is misled by this semantic similarity, thereby failing to effectively distinguish genuine tampering traces from normal, visually similar objects.

Effectiveness of NFIM. The NFIM targets the intrinsic discrepancy between shallow and deep features: shallow features naturally preserve boundary and high-frequency details, while deep features capture more global semantics but may introduce irrelevant semantic propagation during feature fusion. The NFIM adopts level-specific interaction strategies to retain fine-grained boundary details from neighboring shallow features and enhance global semantics from neighboring deep features, then integrates them in a controlled manner. Removing the NFIM leads to inferior localization, suggesting that simple fusion is insufficient and that properly designed neighboring interactions are essential to balance detail preservation and semantic consistency, thereby mitigating false activations induced by irrelevant semantics. This can be observed more intuitively in conjunction with the qualitative visualization in Fig. 7(b). In the ablation result without the NFIM, the predicted mask of the tampered region exhibits severe structural defects (highlighted by the red boxes). On the one hand, obvious holes and missed detections appear within the person's torso (upper red box), indicating that without effective enhancement from neighboring deep features, the model fails to maintain the continuity of the global semantics for the manipulated region. On the other hand, the boundaries of the person's legs become extremely blurry and fragmented (lower red box), demonstrating that brute-force feature fusion discards the valuable fine-grained boundaries and high-frequency detail information present in shallow features.

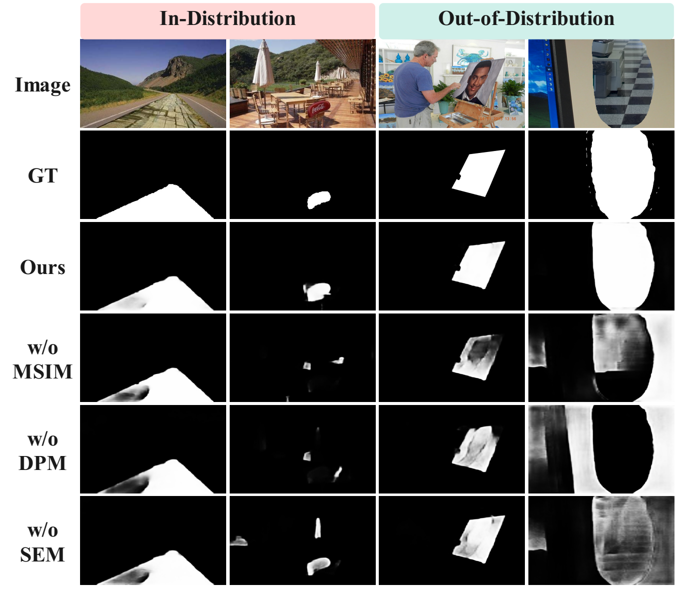

Fig. 8. Visualization of the ablation study on the core components of NFIM, including MSIM, DPM, and SEM.

In addition, we conducted further ablation studies on the internal components of NFIM, namely MSIM, DPM, and SEM. As shown in Table IV, we observed a counterintuitive phenomenon: removing any single component from NFIM resulted in a greater performance drop than removing the entire NFIM. We attribute this primarily to the fact that an incomplete NFIM disrupts the balance of feature fusion across adjacent hierarchical levels, leading to severe confusion and mutual interference among multi-level features. The effectiveness of NFIM relies heavily on the synergy among its constituent components. When cross-level fusion is enforced without an appropriate balancing mechanism, a substantial amount of noise is introduced. This effect can be observed more clearly in the qualitative results shown in Fig. 8, where feature confusion severely degrades the model's localization capability. In some cases, it even causes the model to confuse authentic background regions with tampered areas.

Effectiveness of VFM. The VFM is introduced to incorporate contextual semantic information for validating the predicted masks, thereby improving the reliability of localization masks. In the IML task, when the model relies solely on local suspicious patterns, background regions with textures similar to those of tampered areas may also be over-activated. VFM performs verification-aware fusion by leveraging contextual cues to calibrate and refine the localization results, thereby

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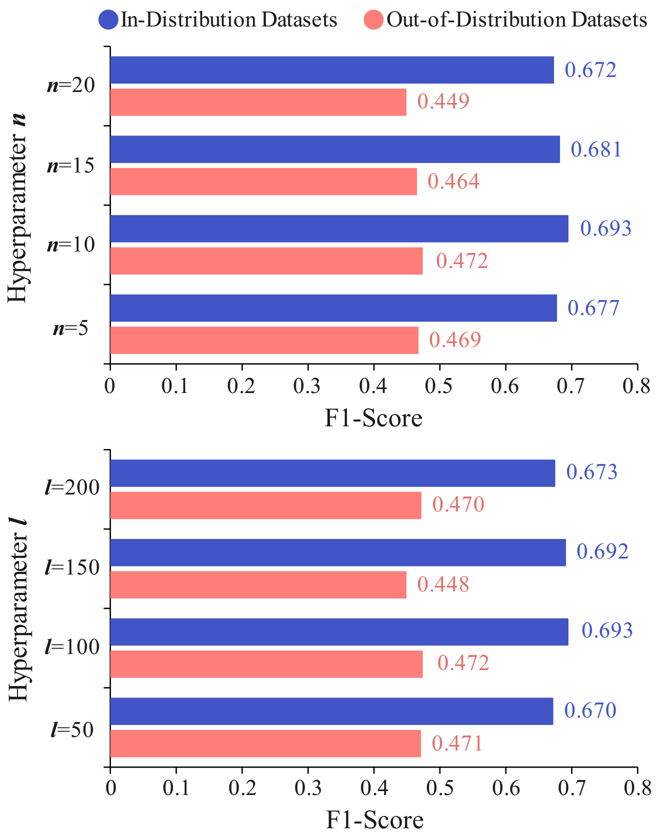

Fig. 9. Performance results across various settings of the memory bank hyperparameters: the number of semantic units (n) and prototype units per cluster (l).

effectively reducing both missed detections and false positives. As shown in Table III, the significant performance degradation caused by removing VFM further underscores its importance in validating and refining tampered-region predictions.

This effect can be observed more intuitively in the qualitative visualization in Fig. 7(c), which further demonstrates VFM's strong capability to correct false alarms in background regions. Without context-aware verification, the model is prone to producing evident false activations in complex backgrounds. By introducing VFM, the model exploits global semantic cues to effectively suppress these distracting responses and accurately correct misclassified regions, thereby substantially improving the accuracy and robustness of IML.

F. Sensitivity Analysis of Memory Bank Hyperparameters

To comprehensively evaluate the robustness of the prototype memory bank in our proposed PNF-Net, we conduct a sensitivity analysis on two core hyperparameters: the number of semantic units (n) and the number of prototype units per cluster (l). The experimental results evaluating the F1-score on both in-distribution (ID) and out-of-distribution (OOD) datasets are illustrated in Fig. 9.

Impact of semantic units (n): The hyperparameter n determines the capacity to represent distinct tampering trace clusters. As shown in the figure, the localization performance initially improves as n increases from 5 to 10, peaking at $n = 10$ with F1-scores of 0.693 on ID datasets and 0.472 on OOD datasets. However, further increasing n to 15 and 20 leads to a steady performance decline. This trend suggests that a severely limited number of semantic units fails to capture the highly diverse patterns of varied image manipulations. Conversely, an excessively large n introduces redundancy and fine-grained noise into the memory space. This causes the model to over-segment the tampering traces into fragmented semantic spaces, ultimately degrading its generalization capability on unseen data.

Impact of prototype units per cluster (l): The hyperparameter l governs the capacity of each semantic cluster to accommodate intra-class variations. Similar to the trend observed with n, the F1-scores consistently reach their maximum at $l = 100$ for both ID and OOD scenarios. When l is set too small ($l = 50$), the memory bank lacks sufficient capacity to robustly represent the rich structural details of local anomalies, yielding sub-optimal F1-scores. However, when l exceeds 100, particularly at $l = 150$ and $l = 200$, the performance drops. This indicates that an overly large pool size per cluster may cause the network to overfit to specific training artifacts, thereby hindering the memory priors from guiding the model's representation learning.

As a result, we empirically set $n = 10$ and $l = 100$ as the optimal default configuration for all subsequent experiments.

G. Robustness Evaluation

To evaluate robustness under common real-world degradations and post-processing, we apply three representative perturbation families to the same test split with multiple severity levels, including additive Gaussian noise, Gaussian blur with kernel sizes, and JPEG compression with quality factors. We report the F1 score for each setting and the average performance across all perturbations (Table V). Overall, our method achieves the highest average F1, surpassing the strongest competing approach by a clear margin, which indicates that PNF-Net maintains more stable localization under distribution shifts induced by noise, blur, and compression. Under Gaussian noise, performance degrades more gradually as the noise level increases, suggesting reduced sensitivity to stochastic pixel perturbations and greater reliance on manipulation-relevant cues rather than fragile local textures. Under Gaussian blur, even when high-frequency boundaries and fine-grained inconsistencies are substantially suppressed, our method remains consistently best across kernel sizes, demonstrating stronger resilience to detail loss and producing more coherent masks. Under JPEG compression, despite blocking artifacts and quantization noise, our approach achieves the top F1 at all quality factors, indicating an improved ability to distinguish genuine manipulation traces from compression-induced pseudo patterns.

While PNF-Net demonstrates robust generalization across various perturbations, it is noteworthy that its performance advantage is particularly pronounced under JPEG compression compared to Gaussian blur. This discrepancy stems from the inherent properties of the proposed prototype memory mechanism and the specific nature of the degradations. JPEG

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TABLE V
ROBUSTNESS COMPARISON ON CASIAV1 DATASET.

Method	None	Gaussian Noise			Gaussian Blur			JPEG Compression			Avg. F1
		3	7	11	3×3	7×7	11×11	50	70	90	
PSCC-Net	0.485	0.496	0.490	0.489	0.472	0.429	0.328	0.171	0.275	0.427	0.397
Trufor	0.523	0.509	0.485	0.473	0.507	0.448	0.382	0.312	0.393	0.470	0.442
MFI-Net	0.670	0.604	0.566	0.529	0.623	0.579	0.489	0.418	0.534	0.607	0.550
SparseViT	0.483	0.474	0.462	0.437	0.460	0.453	0.422	0.143	0.164	0.338	0.373
Mesorch	0.706	0.675	0.661	**0.663**	0.664	0.630	0.578	0.369	0.511	0.660	0.601
Ours	**0.754**	**0.728**	**0.688**	0.626	**0.711**	**0.689**	**0.683**	**0.559**	**0.647**	**0.725**	**0.673**

TABLE VI
ROBUSTNESS EXPERIMENTS ON ONLINE SOCIAL NETWORKS.

Method	App			
	Facebook	WeChat	WeiBo	WhatsApp
PSCC-Net	0.216	0.204	0.229	0.220
Trufor	0.491	0.426	0.496	0.489
MFI-Net	0.641	0.568	0.620	0.640
SparseViT	0.389	0.257	0.432	0.396
PIM	0.438	0.308	0.465	0.463
Mesorch	0.665	0.567	0.647	0.683
Ours	**0.729**	**0.644**	**0.718**	**0.735**

compression introduces structured, high-frequency artifacts. Manipulated regions often exhibit distinct grid mismatches or double-compression inconsistencies, which the memory bank is highly adept at clustering and recalling as stable anomaly priors. In contrast, Gaussian blur acts as a low-pass filter, physically obliterating high-frequency tampering traces such as sharp splicing boundaries and local noise discrepancies. Consequently, while the memory bank still attempts to retrieve relevant prototypes, the severely smoothed input features lack the discriminative stimuli required to strongly activate the corresponding "grandmother cells", leading to a relatively narrowed performance margin against competing methods in highly blurred scenarios.

Furthermore, to emulate unknown post-processing and re-encoding effects in real-world dissemination pipelines, we follow the protocol of MVSS-Net [47] and evaluate localization performance using images that have been compressed by major social platforms, including Facebook, WeiBo, WeChat, and WhatsApp. This pipeline introduces compound degradations such as platform-specific resampling, recompression, and format conversion, which better reflects practical image sharing and reposting scenarios. As shown in Table VI, PNF-Net maintains stable localization performance and produces coherent masks after social-platform processing, further demonstrating its robustness in real deployment settings.

These robustness gains are consistent with our design motivations. The memory-guided priors encourage a preference for manipulation traces and reduce sensitivity to superficial appearance variations. The neighboring feature interaction preserves shallow details while enhancing deep semantic consistency, mitigating degradation-induced feature mismatch.

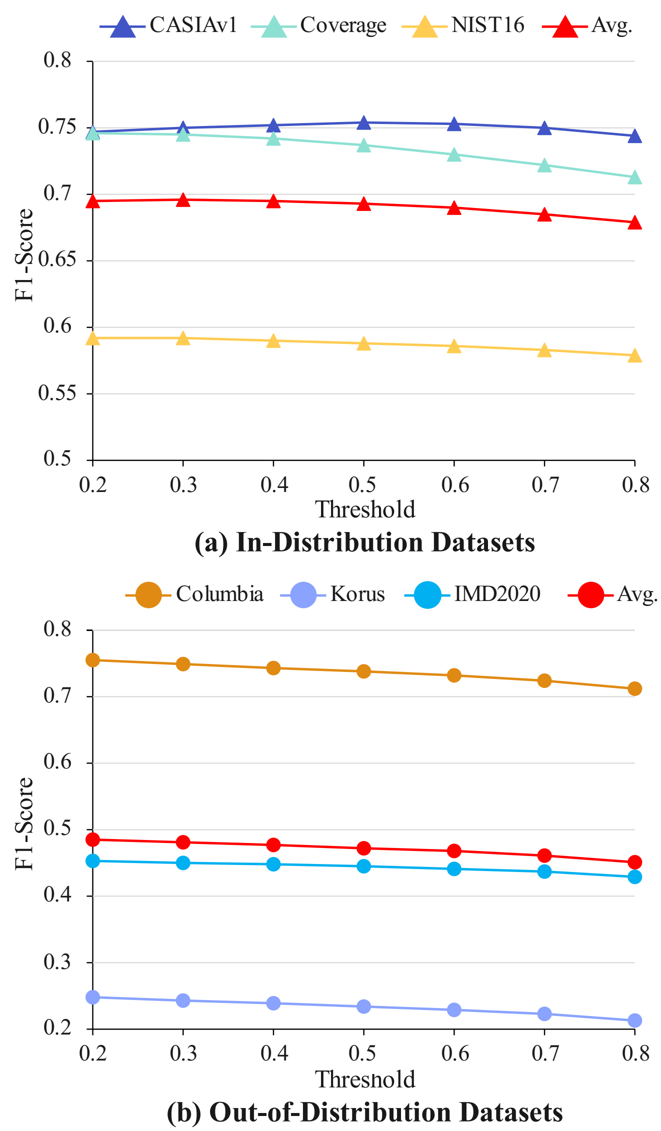

Fig. 10. Impact of threshold variations on the F1 score for (a) in-distribution datasets and (b) out-of-distribution datasets.

The verification fusion further exploits contextual cues to calibrate predictions and complete missing regions, enabling more reliable manipulation localization under both synthetic perturbations and real platform processing.

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H. Threshold Stability Analysis of F1-Scores

As illustrated in Fig. 10, we comprehensively evaluate the F1-score performance of the proposed model across a wide threshold interval. It can be observed that, with the exception of a very slight downward trend on the Coverage dataset, the performance curves across all other in-distribution and out-of-distribution datasets maintain a high degree of flatness. No drastic performance fluctuations occur throughout this extensive threshold variation. This phenomenon strongly indicates that our method is highly immune to changes in classification thresholds. This excellent threshold stability fundamentally stems from the high confidence and feature discriminability of the model's output. The model can accurately separate tampered features from pristine features, enlarging the distance between positive and negative samples in the feature space. This ensures that there are extremely few pixels with predicted probabilities falling into the ambiguous interval. Consequently, sliding the binarization threshold anywhere within this range hardly alters the final predicted mask.

In real-world IML applications, exhaustive threshold fine-tuning for each new scenario is often impractical due to unknown data distributions. The characteristic demonstrated by our model breaks the over-reliance on specific hyperparameters seen in traditional methods. It proves the model's high reliability and robustness against diverse tampering techniques, ensuring stable and highly generalizable detection performance in open-world scenarios.

V. Conclusions

In this paper, we introduce prototype memory technology to the image manipulation localization (IML) field for the first time. By emulating the working mechanism of "grandmother cells" in the primary visual cortex, we construct a prototype memory bank to explore the consistencies and differences between various tampering traces and the background. We guide IML representation learning by finding the most similar prototypes in the memory bank as memory priors. Additionally, we propose a neighboring feature fusion framework to address the issue of irrelevant semantic propagation caused by bottom-up feature fusion. Extensive experiments show that the proposed prototype memory-based neighboring feature fusion network (PNF-Net) significantly outperforms existing SOTA models. Despite these promising results, our approach has certain limitations. Since the prototype memory bank is constructed based on the trace distribution of the training data, PNF-Net may experience degraded performance when confronting completely unseen tampering techniques or extreme image degradations that obscure local trace patterns. Furthermore, the retrieval process within a comprehensive memory bank can introduce additional computational overhead. In future work, we plan to explore dynamic memory updating mechanisms to enhance the model's generalization capabilities against novel tampering methods, and further optimize the memory query architecture to facilitate real-time forensic applications. In summary, our work paves the way for the application of prototype memory technology in the IML field and provides new insights for feature fusion of backbone networks.

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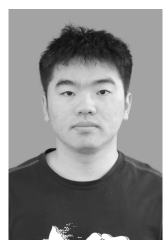

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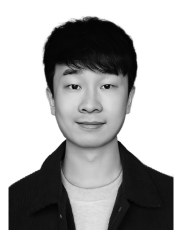

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